Intro Assignment 1

Secure Software & Hardware Systems

Background

"A new employee at EvilCorp believes firmly in new technologies, and now uses ChatGPT to create their programs. You are hired as external consultant and need to demonstrate that this leads to security issues! To do so, you must write a fuzzing harness, find vulnerabilities, and write a report about your findings."

– *https://canvas.bham.ac.uk/courses/83848/assignments/554062*

The assignment

- Task 1: Fuzzing
  - Fuzz the target code and find a bug!
- Task 2: Vulnerability analysis and patching
  - Patch this bug and demonstrate efficacy.
- Task 3: Reporting
  - Classify this bug with a CVE score.
  - Report your findings.

About Dockerfiles

- Describes the container that we will build and run to evaluate your work.
- For parts one and two, this will contain the majority of your work.
- Prefer strongly to edit the `Dockerfile` than ‘`run.sh`’!
- Don’t forget to include ***everything*** needed for your `Dockerfile` to build and run!
- Consider downloading your submission somewhere else, extracting it, and making sure that it runs by itself.

Building AFL++

- Please keep build times under 5 or so minutes.
- This does not include downloading (“pulling”) your base image.
- This will require that you do not build all of AFL++ with ‘`make distrib`’!
- Of course fuzzing should and will take longer than this!

About Patch files

- Patch files describe the ***difference*** between two files.
- You should include a patch that fixes the bug you find.
- We are happy with any format that the ‘`patch`’ tool understands.
- Check ‘`patch(1)`’ with ‘`man patch`’!

Submission Timeline

- Assignment available:
  - 29 Oct 2025, 09:00

- Submission closes
  - 17 Nov 2025, 15:59

- Late submissions:
  - 5% penalty for each day late

Plagiarism

- We take plagiarism seriously.
    - We expect that your submissions reflect the work of your group
    - And that the submission is formulated in **your own words**

- Do not copy code/text from third parties without referencing it
    - This includes submitting generative AI generated texts/code as your own work
    - This includes **also** the use of generative AI based translation tools:
        - *“In particular, generative AI or other editorial assistance must not be used to: [...] Translate text into, or from, the language being studied.”*

– UoB’s Code of Practice on Academic Integrity, A.1.6 [link]

On GenAI (1)

GenAI: Unacceptable proof-reading / editorial assistance

Rewriting or editing of text with the purpose of improving the Student's research arguments or contributing new arguments or rewriting computer code is not acceptable, whether undertaken by a person, by generative AI or by any other means, and may be deemed to be plagiarism. In particular, generative AI or other editorial assistance must not be used to:

- *Alter text to clarify and/or develop the ideas, arguments, and explanations.*
- *Correct the accuracy of the information.*
- *Develop or change the ideas and arguments presented.*

UoB's Code of Practice on Academic Integrity, A.1.6 [link]

On GenAI (2)

Generative AI or other editorial assistance may only be used to offer advice and guidance on:

• Correcting spelling and punctation.

• Ensuring text follows the conventions of grammar and syntax in written English.

• Shortening long sentences or reducing long paragraphs without changes to the overall content.

• Ensuring the consistency, formatting and ordering of page numbers, headers and footers, and footnotes and endnotes.

• Improving the positioning of tables and figures and the clarity, grammar, spelling and punctuation of any text in table or figure legends.

UoB's Code of Practice on Academic Integrity, A.1.6 [link]

TA Support

If you have questions, please reach out to all three TAs via email:

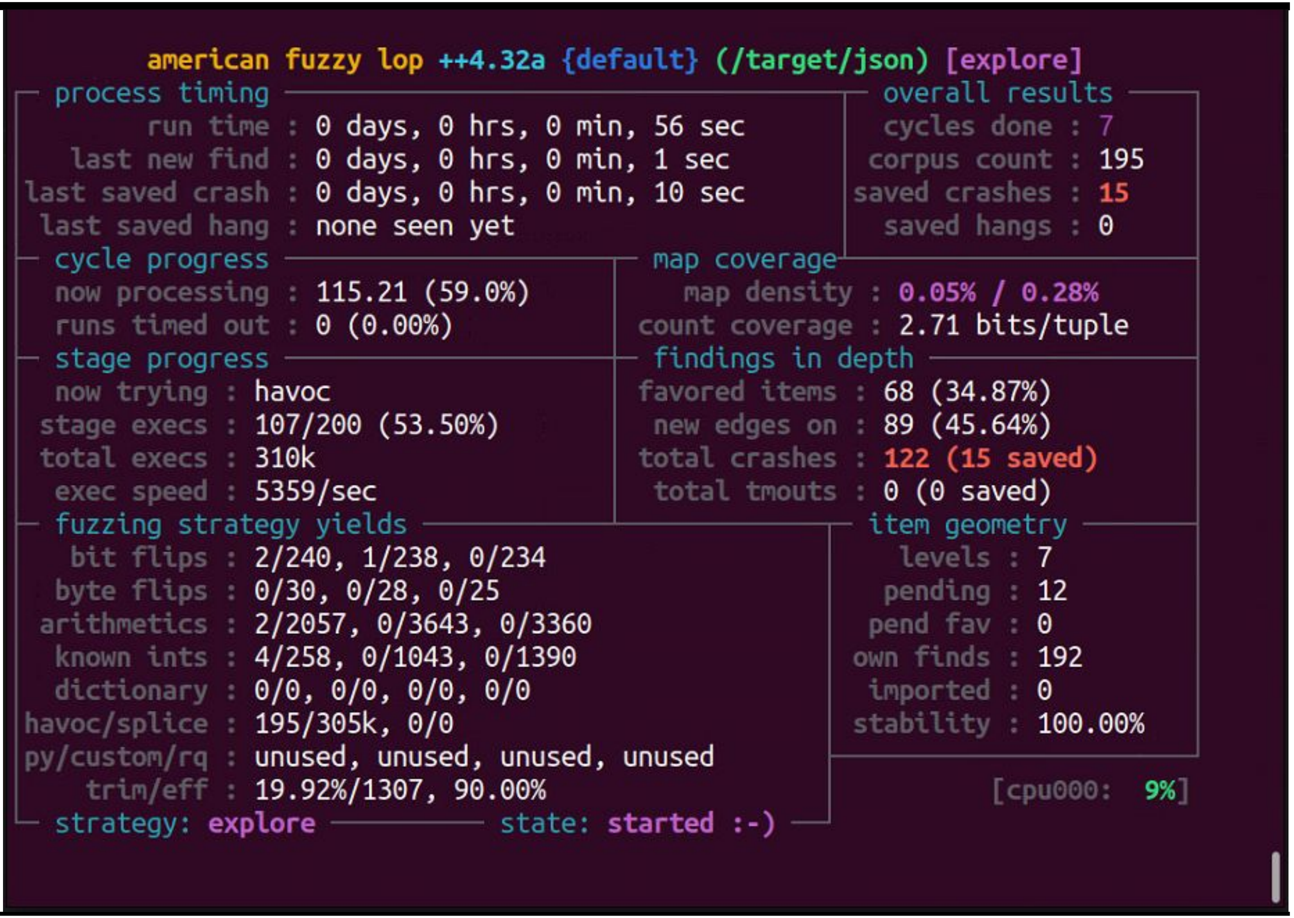
- Jinjin Wang <jxw1663@student.bham.ac.uk>
- Yusuf Hegazy <yah408@student.bham.ac.uk>

- Make sure to include **all two** in your emails
- And when answering, to use **reply all**

During reading week, TA support will be available during the lecture slots:

- Monday, Nov 3rd, 12.00-13.00: CS, 117
- Tuesday, Nov 4th, 10.00-11.00: CS, 117

Questions?


american fuzzy lop ++4.32a {default} (/target/json) [explore]
process timing
run time : 0 days, 0 hrs, 0 min, 56 sec
last new find : 0 days, 0 hrs, 0 min, 1 sec
last saved crash : 0 days, 0 hrs, 0 min, 10 sec
last saved hang : none seen yet
overall results
cycles done : 7
corpus count : 195
saved crashes : 15
saved hangs : 0
cycle progress
now processing : 115.21 (59.0%)
runs timed out : 0 (0.00%)
map coverage
map density : 0.05% / 0.28%
count coverage : 2.71 bits/tuple
stage progress
now trying : havoc
stage execs : 107/200 (53.50%)
total execs : 310k
exec speed : 5359/sec
findings in depth
favored items : 68 (34.87%)
new edges on : 89 (45.64%)
total crashes : 122 (15 saved)
total tmouts : 0 (0 saved)
fuzzing strategy yields
bit flips : 2/240, 1/238, 0/234
byte flips : 0/30, 0/28, 0/25
arithmetics : 2/2057, 0/3643, 0/3360
known ints : 4/258, 0/1043, 0/1390
dictionary : 0/0, 0/0, 0/0, 0/0
havoc/splice : 195/305k, 0/0
py/custom/rq : unused, unused, unused, unused
trim/eff : 19.92%/1307, 90.00%
item geometry
levels : 7
pending : 12
pend fav : 0
own finds : 192
imported : 0
stability : 100.00%
strategy: explore
state: started :-)
[cpu000: 9%]